

PROJECT REPORT

Visualization Tool for Electric Vehicle Charge and Range Analysis

1. INTRODUCTION

1.1 Project Overview

The Visualization Tool for Electric Vehicle Charge and Range Analysis is a data visualization project developed using MySQL and Tableau. The project analyzes electric vehicle datasets containing information about EV models, charging stations, vehicle range, efficiency, pricing, top speed, body style, and powertrain. By transforming raw data into interactive dashboards and charts, the system helps users understand EV trends and make informed decisions.

The project integrates multiple datasets stored in a MySQL database and visualizes them through Tableau dashboards. Interactive filters, calculated fields, maps, bar charts, treemaps, and summary cards provide meaningful insights into electric vehicle performance and charging infrastructure.

1.2 Purpose

The main purpose of this project is to simplify the analysis of electric vehicle data using interactive visualizations. The objectives are:

To analyze electric vehicle specifications and charging infrastructure.

To visualize EV trends using Tableau dashboards.

To compare EV brands based on range, efficiency, speed, and pricing.

To display charging station distribution across India.

To support users in making better EV-related decisions through data-driven insights.

. Below is the remaining report content you can use.

2 Perfect. IDEATION PHASE

2.1 Problem Statement

The increasing adoption of electric vehicles has generated a large amount of data related to vehicle performance, charging stations, range, efficiency, and pricing. However, this information is often scattered across multiple sources, making it difficult for users to analyze and compare EVs effectively. There is a need for an interactive visualization system that presents EV data in a simple, meaningful, and user-friendly format.

2.2 Empathy Map Canvas

Says

- Need reliable EV information.
- Compare vehicle prices and range.
- Find charging stations easily.

Thinks

- Which EV offers the best value?
- Is the charging infrastructure sufficient?
- Which brand is most efficient?

Does

- Searches EV specifications online.
- Compares different brands.
- Reviews charging station availability.

Feels

- Confused due to scattered information.
- Wants accurate and reliable insights.
- Needs a simple visualization platform.

2.3 Brainstorming

Different ideas were discussed before selecting the final project. The team identified that data visualization could simplify complex EV data. Tableau dashboards combined with MySQL storage were selected because they provide interactive analysis and easy understanding of trends.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer begins by searching for EV information. The system allows users to view vehicle details, compare brands, analyze charging stations, apply filters, and explore dashboards to gain meaningful insights before making decisions.

3.2 Solution Requirements

Functional Requirements

- User can access EV dashboards.
- User can compare EV brands.
- User can view charging station locations.
- User can filter data by brand and charging type.
- User can analyze range, efficiency, speed, and pricing.
- User can interact with dashboard visualizations.

Non-Functional Requirements

- Easy to use interface.
- Secure database connectivity.
- Reliable data visualization.
- Fast dashboard performance.
- High availability.
- Scalable architecture.

3.3 Data Flow Diagram

The system collects EV datasets, stores them in MySQL, processes the data, connects Tableau to the database, generates interactive visualizations, and presents insights to users through dashboards and stories.

3.4 Technology Stack

Layer	Technology
Front End	Tableau Dashboard
Database	MySQL
Data Visualization	Tableau Desktop
Programming	SQL
Operating System	Windows
Dataset Source	Skill Wallet

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The project addresses the difficulty of analyzing EV information by providing an interactive dashboard that combines vehicle specifications, charging station details, and performance metrics into a single visualization platform.

4.2 Proposed Solution

The proposed solution integrates EV datasets into a MySQL database and connects them with Tableau. Interactive dashboards, maps, summary cards, and charts allow users to compare brands, charging stations, efficiency, prices, and vehicle performance.

4.3 Solution Architecture

The architecture consists of four main layers:

Data Source (EV datasets)

MySQL Database

Tableau Processing Layer

Interactive Dashboard for Users

The system retrieves data from MySQL, processes it in Tableau, and presents meaningful visualizations through dashboards and stories.

5. PROJECT PLANNING & SCHEDULING

The project was developed using Agile methodology.

Major activities included:

Data Collection

Data Preparation

Database Creation

SQL Operations

Tableau Integration

Visualization Development

Dashboard Creation

Story Creation

Performance Testing

User Acceptance Testing

Story points were assigned using the Fibonacci estimation technique and completed through planned sprints.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Data Collection

EV datasets were collected from Skill Wallet and imported into the MySQL database. SQL queries were executed to verify the data before visualization.

6.2 Data Processing

The EV datasets were imported into MySQL and connected to Tableau. The data was reviewed, field names and data types were verified, calculated fields were created where

required, filters were applied for analysis, and the processed data was validated before generating the visualizations.

6.3 Performance Testing

Performance testing confirmed that:

- Database connection was successful.
- Data rendered correctly.
- Interactive filters worked properly.
- Two calculated fields were used.
- Dashboard contained 11 visualizations.
- Story contained 4 story points.

6.4 User Acceptance Testing

Users verified:

- Dashboard accessibility
- Correct data visualization
- Filter functionality
- Map visualization
- Dashboard responsiveness

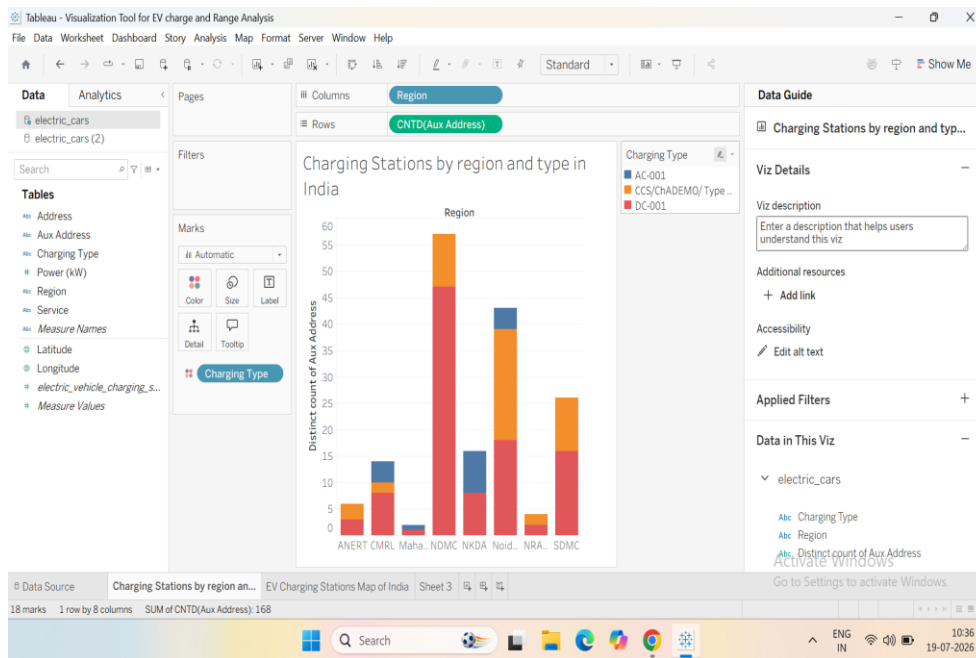
All critical test cases passed successfully.

7. RESULTS

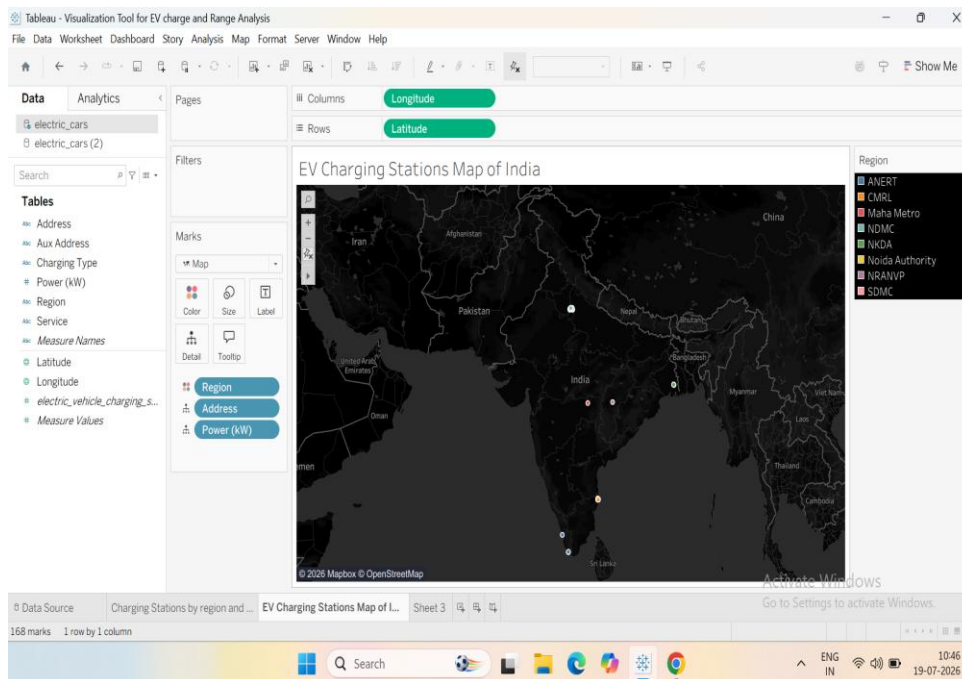
The project successfully generated interactive Tableau dashboards for EV analysis.

Visualizations include:

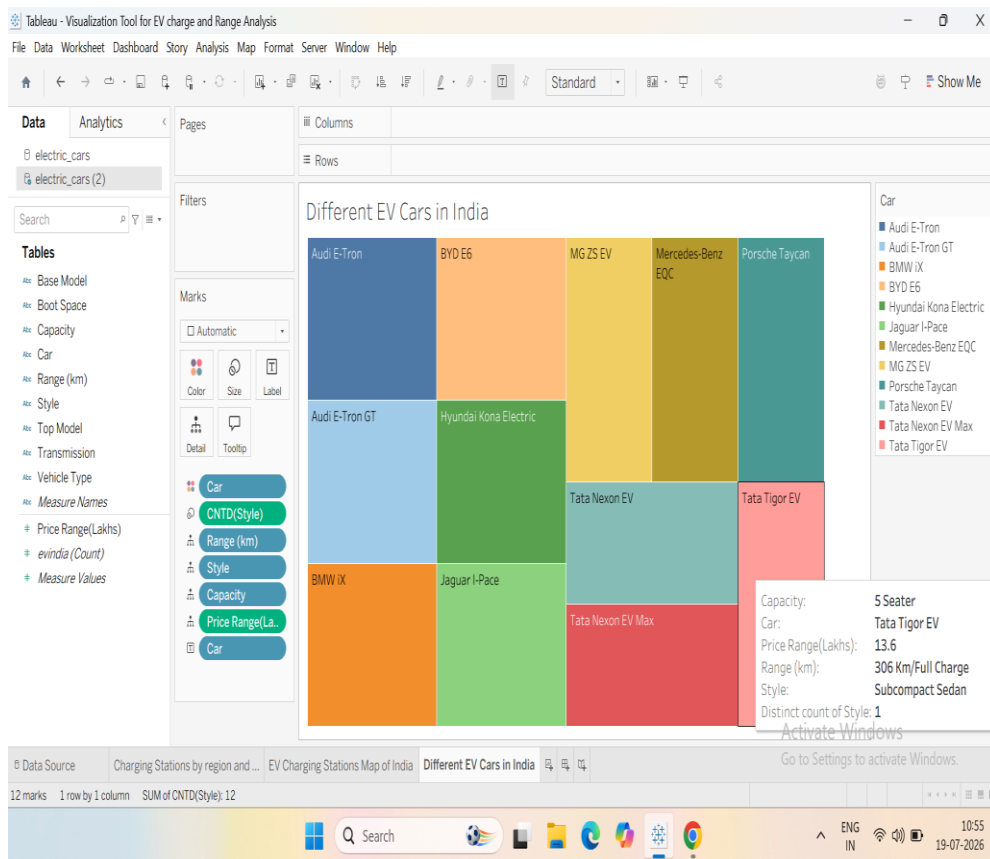
- Charging Stations by Region and Type



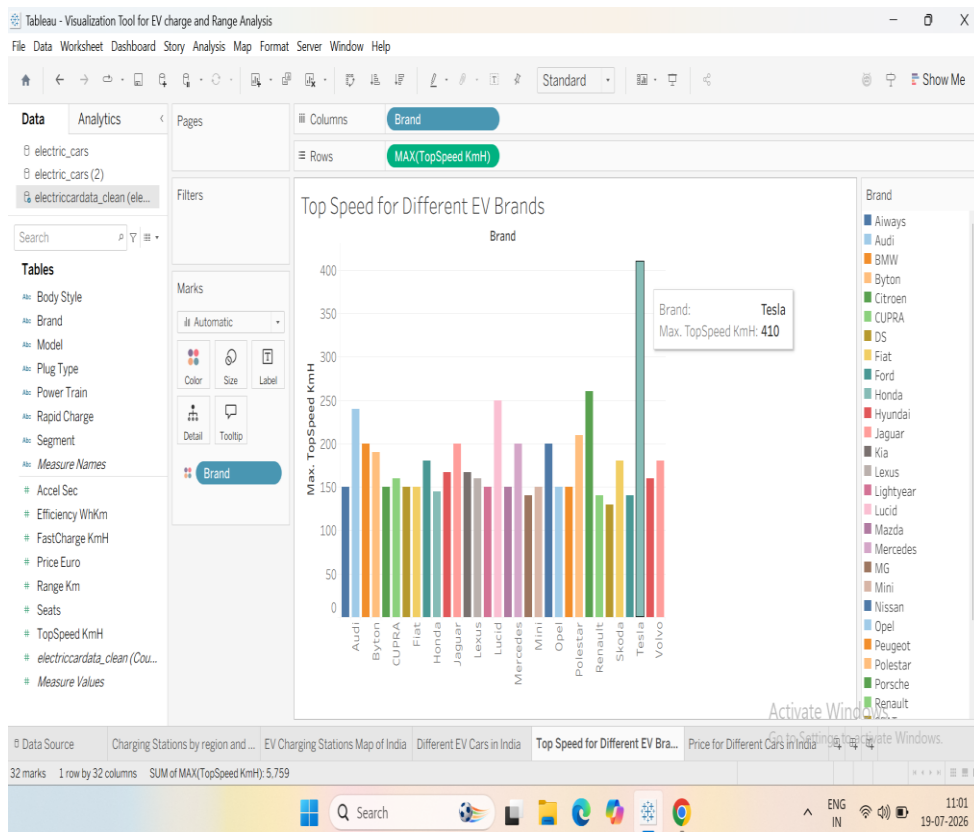
- EV Charging Station Map



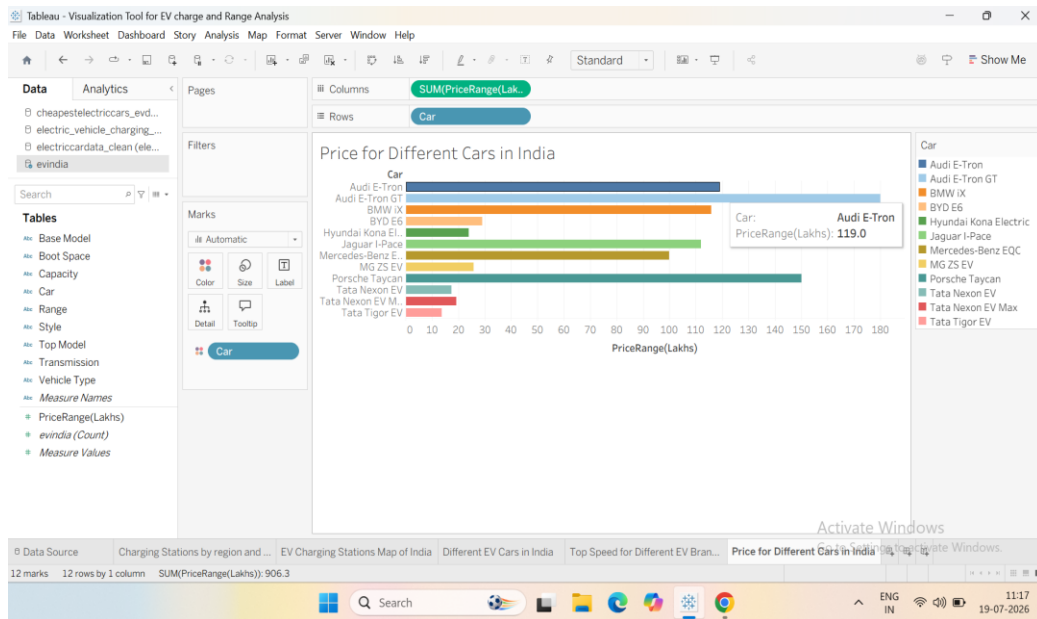
- Different EV Cars in India



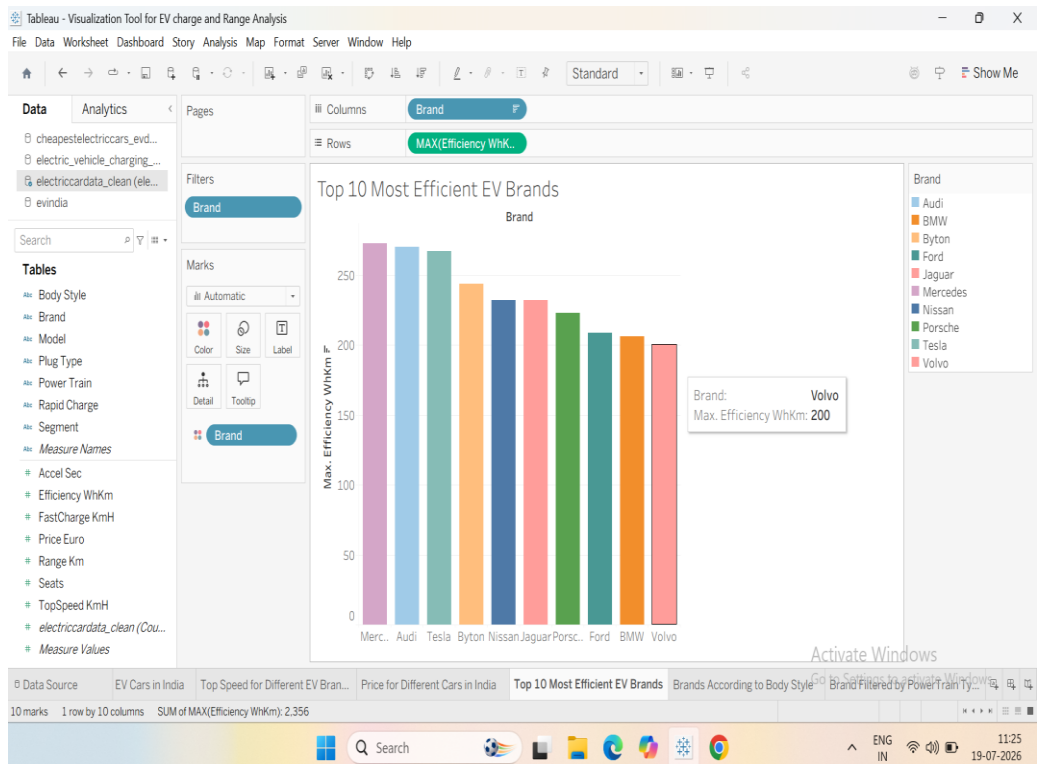
- Top Speed by Brand



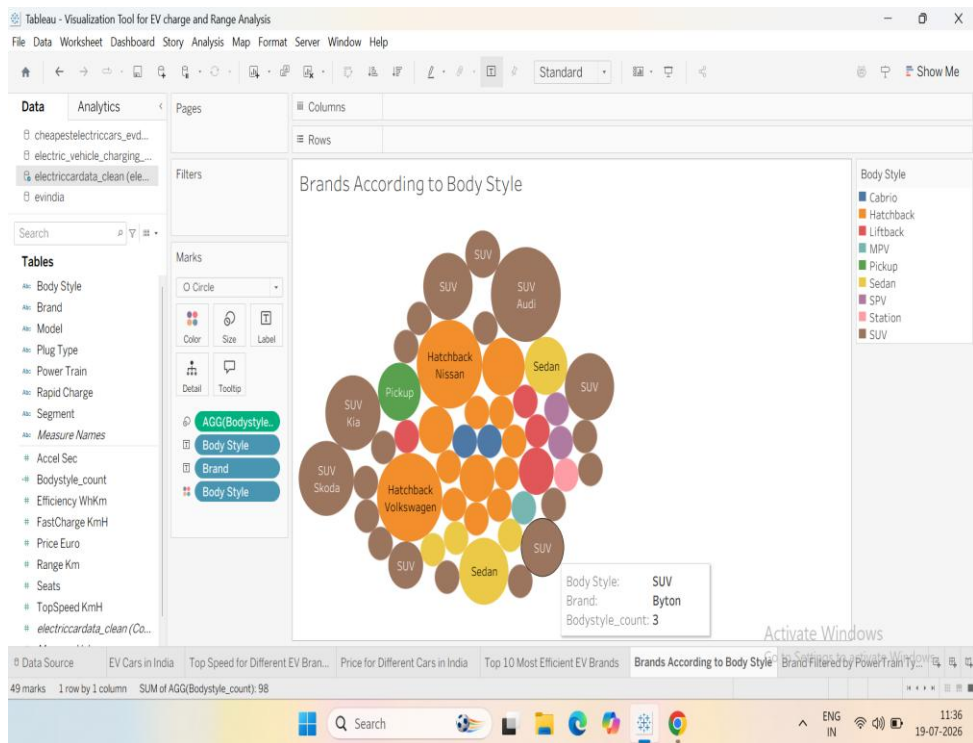
- Price Comparison



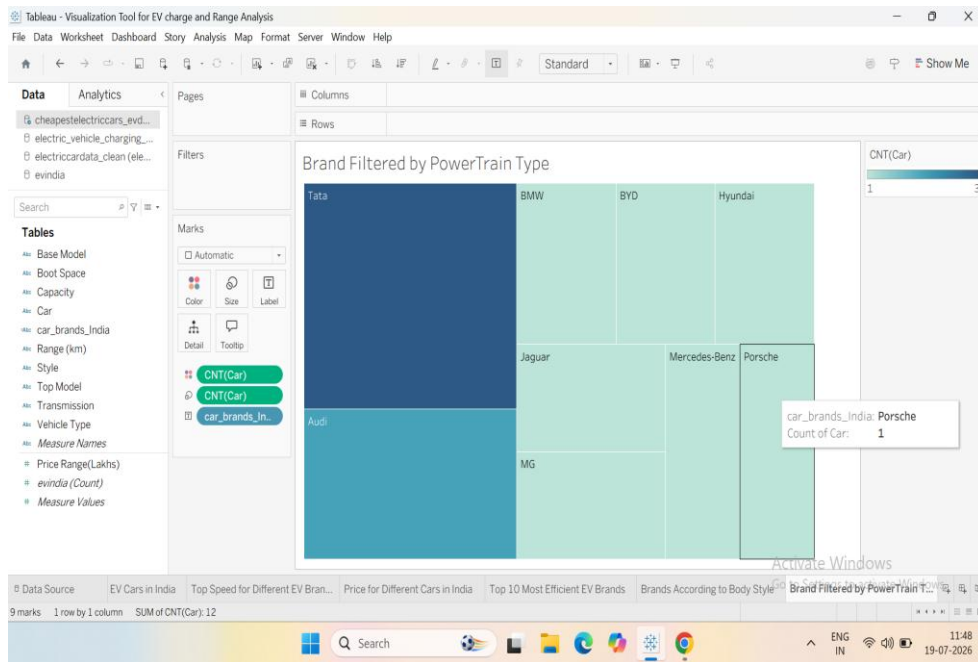
- Top 10 Efficient EV Brands



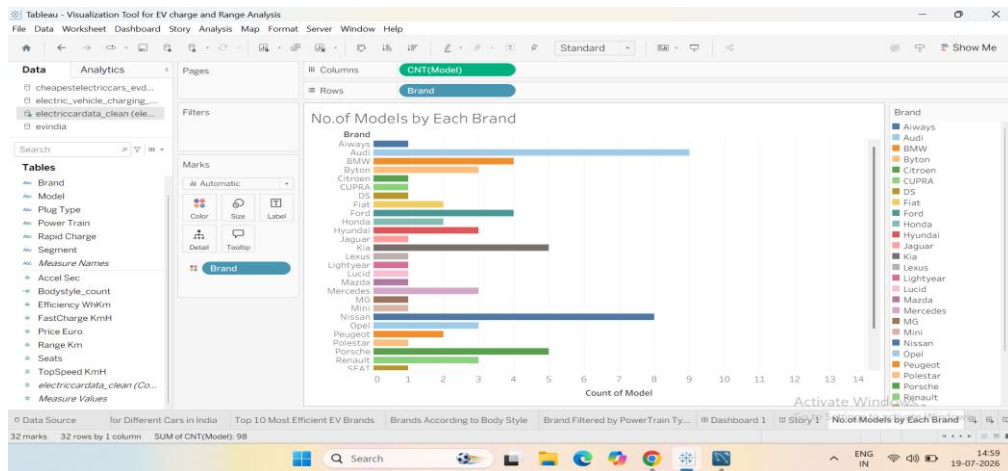
- Brands by Body Style



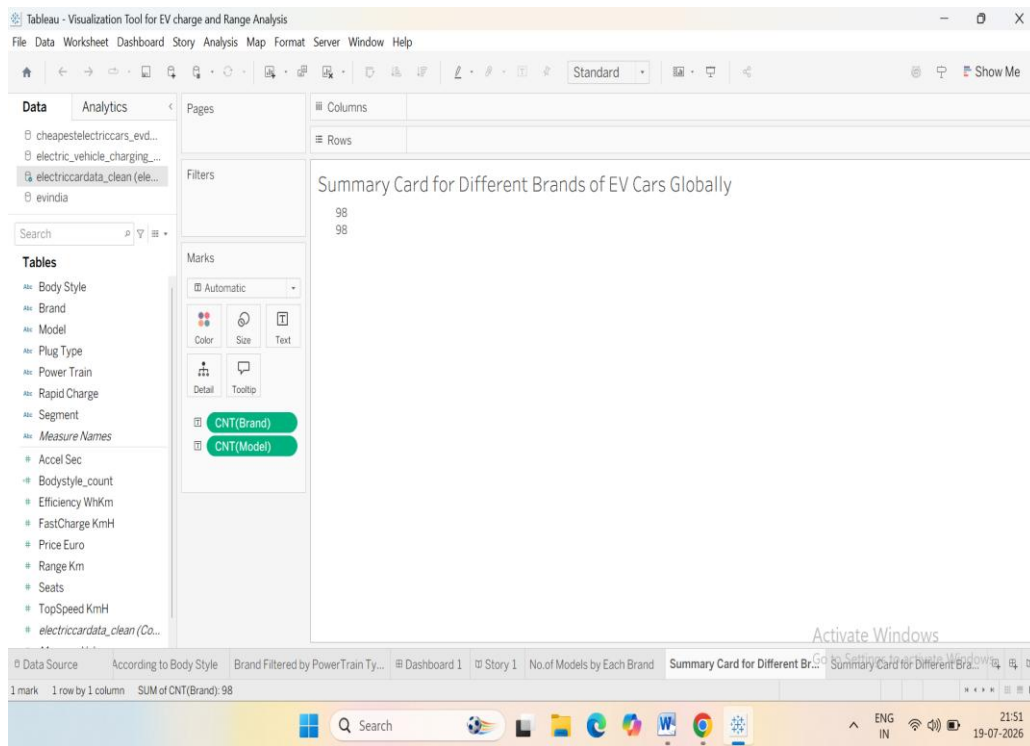
- Brand by Power Train



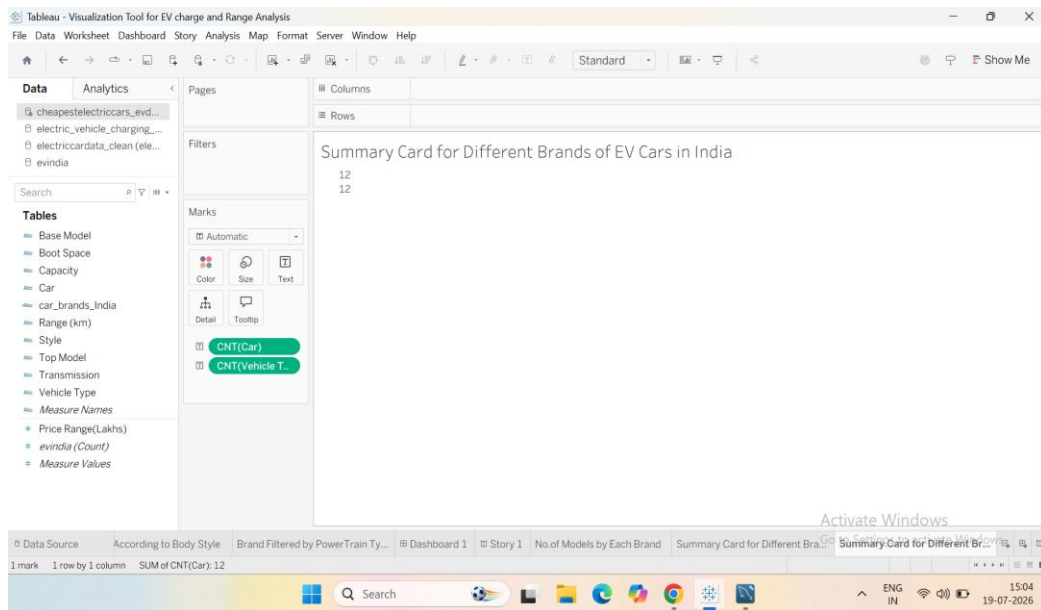
- Number of Models by Brand



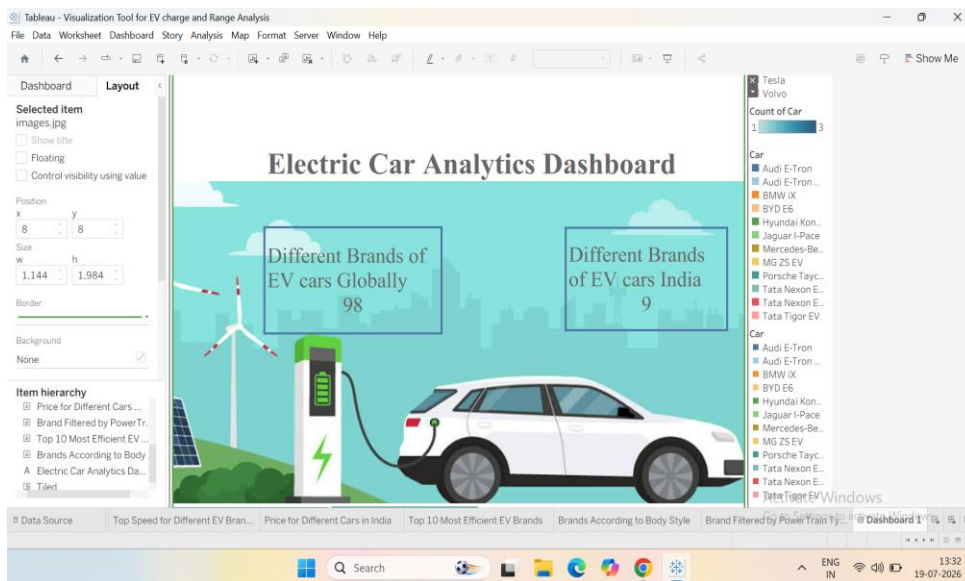
- Global EV Brand Summary Card

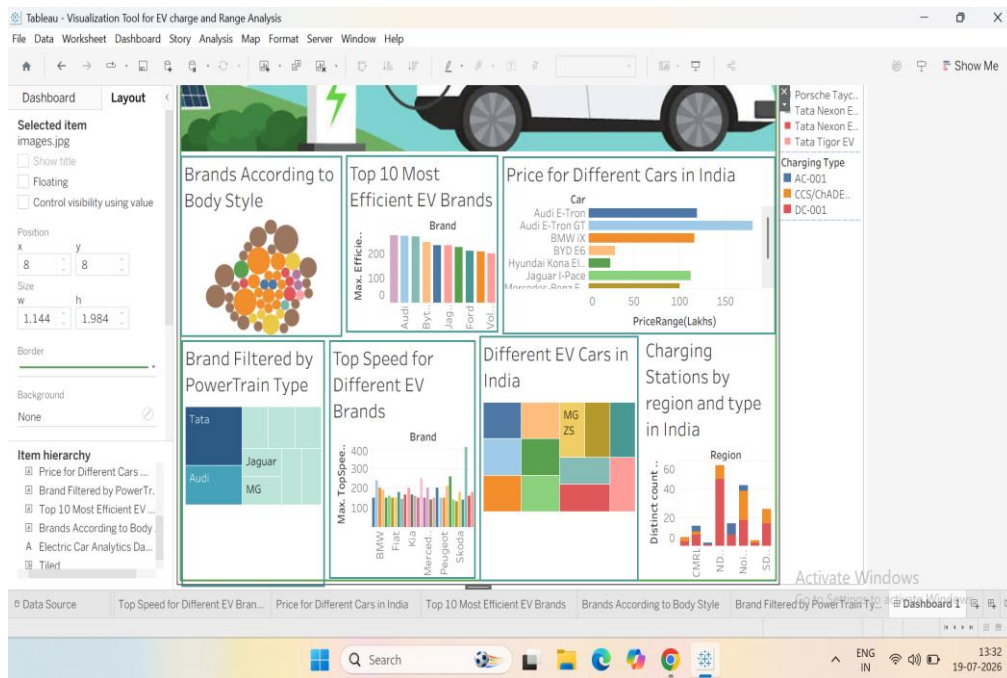


- India EV Brand Summary Card

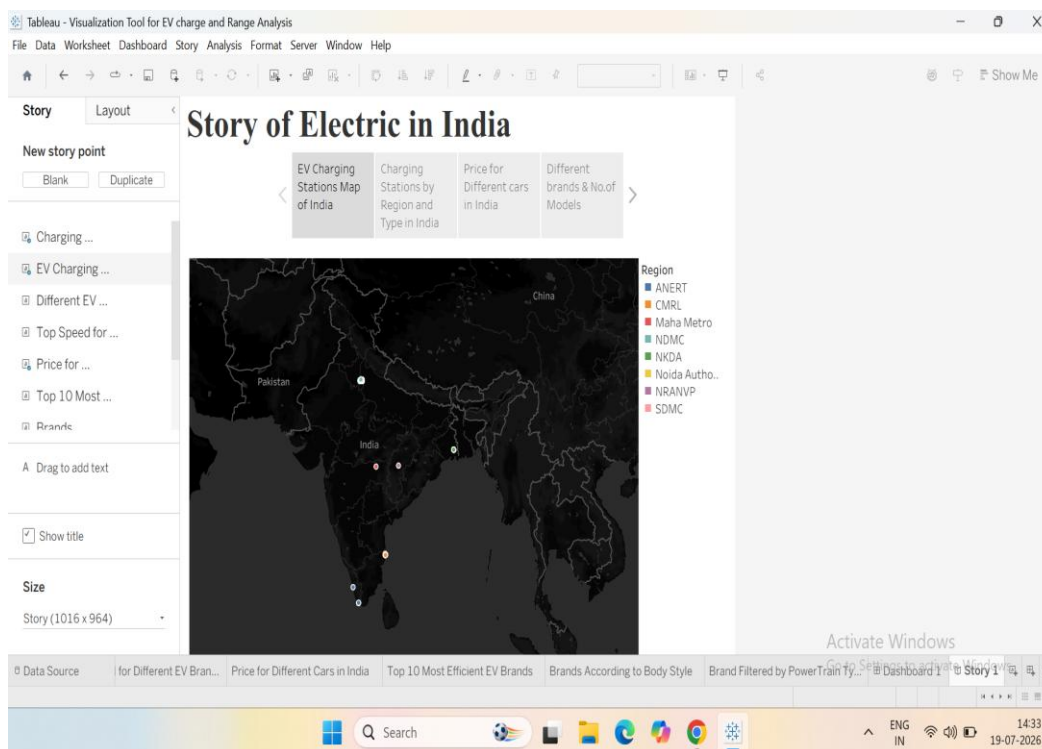


Dashboard





Story



The dashboards provided meaningful insights into EV performance, charging infrastructure, and vehicle comparison.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive dashboard
- Easy comparison of EV brands
- Fast data analysis
- User-friendly interface
- Supports decision-making
- Attractive visualizations

Disadvantages

- Depends on dataset quality
- Requires Tableau Desktop
- Data must be updated manually
- Limited to available datasets

9. CONCLUSION

The Visualization Tool for Electric Vehicle Charge and Range Analysis successfully transforms raw EV data into interactive dashboards using MySQL and Tableau. The project enables users to analyze charging stations, compare vehicle specifications, evaluate efficiency, and gain valuable insights through visual analytics. The developed system meets the project objectives and demonstrates the effectiveness of data visualization for electric vehicle analysis.

10. FUTURE SCOPE

Future enhancements include:

Live data integration using APIs.

Real-time charging station availability.

Predictive analytics using Machine Learning.

Mobile dashboard support.

Cloud deployment.

AI-based EV recommendation system.

11. APPENDIX

Source Code: SQL scripts and Tableau Workbook (.twb/.twbx)

GitHub Repository: <https://github.com/Poojitha-chitti/Visualization-Tool-for-Electric-Vehicle-Charge-and-Range-Analysis>

Project Demonstration: Demo video link